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ECHO: PRUNE TO ACT, TRACE TO LEARN WITH SELECTIVE TURN MEMORY IN AGENTIC RL

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ABSTRACT

Long-horizon language agents must repeatedly interact with tools, accumulate evidence, and make decisions under bounded context windows. Existing context-management methods make such rollouts feasible by truncating distant history, folding past turns into summaries, or selecting compact memory states. However, these breakthroughs introduce two coupled limitations. First, as the number of turns grows, historical observations are progressively removed or collapsed into compressed states, making it harder for the policy to reuse fine-grained evidence. Second, once the original turns are no longer source-addressable, outcome-based RL loses an explicit path for aligning policy updates with the evidence that supported a successful final answer. To this end, we propose ECHO, a selective turn-memory framework that jointly addresses history collapse and traceable learning through source-indexed reconstruction. Specifically, ECHO compresses each completed environment turn into a compact memory record, reconstructs bounded policy contexts by selecting from these records, and reuses the selected source indices to route positive outcome credit to the evidence and selection actions that support successful answers. On BrowseComp-Plus, ECHO reaches 43.4% held-out accuracy, outperforming GRPO (28.9%) and the rolling-summary baseline SUPO (36.1%), while using fewer turns and lower trajectory volume than SUPO (Figure 1). Additionally, the trained policy improves zero-shot generalization across multi-objective QA, code generation, and deep information-seeking benchmarks on both dense and MoE backbones.

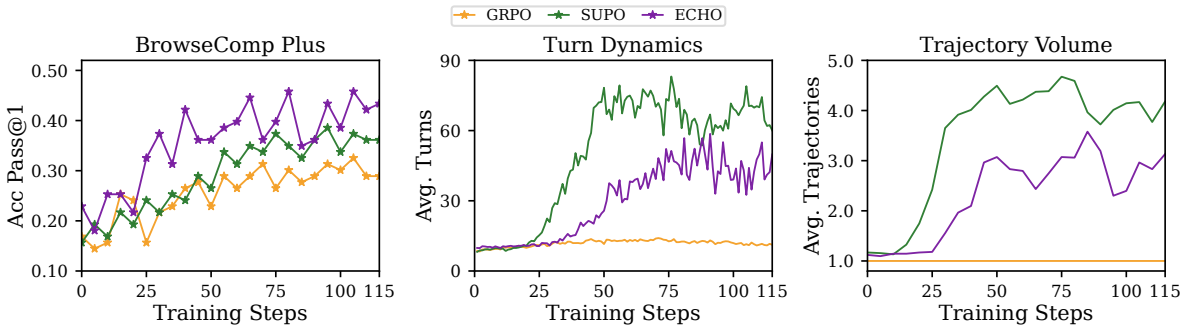


Figure 1: Held-out accuracy, tool-use turns per rollout, and trajectory volume over training on BrowseComp-Plus with the Qwen3-32B-Instruct backbone for ECHO (purple), GRPO (orange), and SUPO (green). ECHO traces the upper-left frontier: rising accuracy without the turn and volume growth seen for SUPO.

1 INTRODUCTION

Large language models (LLMs) increasingly act as multi-turn agents that interleave reasoning, tool invocation, and environment feedback (Yao et al., 2023; Schick et al., 2023). Recently, reinforcement learning (RL) from verifiable final outcomes has become a central recipe for improving such agents in search, coding, function calling,

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and deep-research settings (Jin et al., 2025; Qian et al., 2025; Li et al., 2025; Zheng et al., 2025). However, as horizons grow, interaction history becomes a bottleneck for both acting and learning. The agent needs to retain useful tool observations within a bounded context window, while the learner must identify which earlier turns should be reinforced from the final outcome signal.

Context-management methods address the acting bottleneck by truncating, summarizing, retrieving, or editing history before the next decision (Li et al., 2024; Packer et al., 2023; Kang et al., 2025). These operations make long rollouts feasible, but they introduce two coupled limitations. First, as the number of turns grows, distant observations are removed, compressed, or folded into collapsed states, making fine-grained evidence harder to reuse. Second, once the original turns are no longer source-addressable, outcome-based RL lacks an explicit route for aligning policy updates with the evidence turns that later decisions relied on. This issue is especially pronounced in collapsed-state methods such as rolling summarization, where the policy may condition on information derived from earlier observations, but the learner only sees the summary-generation tokens rather than the original evidence sources (Lu et al., 2025; Yu et al., 2026).

The lack of source-level addressability directly affects how sparse outcome rewards are used for training. With final-outcome rewards, standard multi-turn RL often applies a trajectory-level advantage to many generated spans (Schulman et al., 2017; Shao et al., 2024). When the reconstructed context is not source-addressable, this dense assignment can reinforce redundant searches, incidental reasoning, or summary-generation behavior together with genuinely useful evidence-gathering steps. Our diagnostics show that a rolling-summary baseline can extend rollouts, but also leads to turn proliferation, larger trajectory volume, and slower training. Thus, making rollouts longer is not sufficient: long-horizon agentic RL also requires a reconstruction trace that records which evidence was selected and reused.

In this paper, we propose ECHO, a selective turn-memory framework for traceable context reconstruction in agentic RL. After each completed tool-use turn, ECHO writes a compact memory record containing a local finding and a source-turn pointer. Distant history is maintained as a non-collapsing set of such records. When the context budget becomes binding, the policy selects useful memory records and reconstructs the next bounded context from those records plus recent interactions. The agent therefore acts with compact context, while the selected memories remain linked to their original environment turns.

The same source indices also guide learning. Instead of spreading outcome advantages over every generated token, ECHO routes positive outcome credit through the final answer segment, selected historical source turns, their memory findings, and the memory-selection actions that constructed the reconstructed context. In this way, ECHO aligns context pruning for acting with outcome-credit routing for learning through a single source-indexed reconstruction trace. We formalize this idea through a context-managed multi-turn RL interface, under which different context managers can be compared by whether reconstructed information remains addressable at the level of original environment turns. Our main contributions are as follows:

- We formulate context-managed multi-turn RL with a unified reconstruction interface that separates history state, bounded-context reconstruction, and source-level traceability. This view makes explicit how collapsed-history methods can preserve context efficiency while losing the provenance needed for outcome-credit assignment.
- We introduce ECHO, a selective turn-memory reconstruction method that stores each completed turn as a source-indexed memory and constructs bounded policy contexts through learned memory selection rather than global history collapse.
- We propose provenance-guided token-level credit assignment, which reuses selected source indices to route positive outcome advantage to final answers, selected historical evidence turns, memory findings, and memory-selection actions.
- Experiments on BrowseComp-Plus and diverse zero-shot benchmarks show that ECHO improves accuracy while reducing turn proliferation and trajectory volume relative to rolling-summary training, with consistent gains across dense and MoE backbones.

2 Related Work

2.1 RL for Long-Horizon Language Agents

Language agents are often formulated as multi-turn decision makers that interleave reasoning, tool invocation, and environment feedback (Yao et al., 2023). Recent RL methods optimize such agents with outcome-based or verifiable rewards for search, function calling, coding, and deep-research tasks (Schulman et al., 2017; Shao et al., 2024; Guo et al., 2025; Jin et al., 2025; Song et al., 2025; Li et al., 2025; Qian et al., 2025; Zheng et al., 2025; Du et al., 2025). Long-horizon benchmarks further expose the need for agents that repeatedly search, compare evidence, and verify answers over many turns (Wei et al., 2025; Chen et al., 2025; Qiao et al., 2025).

These settings motivate bounded-context rollouts, but standard outcome-based RL does not specify how managed historical context should expose provenance for credit assignment from final outcome signals.

2.2 Context Management and Agent Memory

A broad line of work addresses long-context limitations through prompt compression, summarization, retrieval, and explicit memory modules (Li et al., 2023; 2024; Wang et al., 2024; Xu et al., 2024; Shen et al., 2025; Yang et al., 2025). Memory-augmented agents maintain external stores that can be written to and retrieved across interactions (Packer et al., 2023; Zhong et al., 2024; Chhikara et al., 2025; Xu et al., 2025; Shan et al., 2025; Wang & Chen, 2025), with recent systems structuring memory through dependency graphs, subgoal-oriented working memory, or optimized compression guidelines (Wu et al., 2026; Hu et al., 2024; Kang et al., 2025). Recoverable-memory methods such as SAM keep raw trajectory pages accessible through compact memory cues and recall them when needed (Hu et al., 2026). These methods improve acting under limited context, but retrieval or compression alone does not define how final outcome signals should be linked back to original source turns.

Recent work has begun to make context management part of agentic RL. Some methods replace the distant prefix with a recursively updated summary, reasoning state, working memory, or compact internal state (Wu et al., 2025; Lu et al., 2025; Yu et al., 2026; Zhou et al., 2025). Other methods treat context management as explicit operations or actions, such as adding, updating, deleting, pruning, or omitting memory items (Yan et al., 2025; Zhang et al., 2025; Ning et al., 2026). Relatedly, MemPO optimizes self-generated memory spans with memory-level advantages (Li et al., 2026). ECHO differs from these approaches by keeping each completed turn source-indexed and reusing the same selected memories that reconstruct the policy context as provenance routes for outcome-credit assignment.

2.3 Credit Assignment in Long-Horizon Agentic RL

Credit assignment is a classical challenge when rewards are sparse and observed only after extended interactions (Williams, 1992; Sutton & Barto, 2018). In many outcome-supervised reinforcement learning methods for language agents, a trajectory- or response-level scalar advantage is reused across generated action tokens or action spans (Schulman et al., 2017; Shao et al., 2024; Yu et al., 2025). This dense assignment is simple, but can be noisy in long-horizon tool-use tasks where only a small subset of past observations supports final success.

Recent work refines credit along temporal, uncertainty-based, or architectural axes. Turn-level methods estimate step-specific advantages for tool-use interactions (Zeng et al., 2025); hindsight methods rescore historical actions using outcome information (Tan et al., 2026); milestone-guided methods partition rollouts at subtask boundaries (Wang et al., 2026); and hierarchical methods separate credit between high-level planning and low-level execution (Peng et al., 2026). Survey work further organizes these methods across token-, step-, turn-, and segment-level credit assignment (Zhang, 2026). Orthogonal to these approaches, uncertainty-based methods such as AEM modulate response-level advantages using entropy-derived signals (Zhao et al., 2026).

These methods improve how final outcome rewards are distributed over time, uncertainty, or policy levels, but they are mostly independent of how the agent's effective context was reconstructed. ECHO is complementary: rather than estimating a new temporal advantage, it uses the source indices exposed by context reconstruction to define a provenance-guided token mask for outcome-credit routing.

3 Preliminaries

3.1 Multi-Turn GRPO for Agentic RL

Let x denote an initial task prompt. At interaction step t, the language agent conditions on a policy context c_t, which contains x and a managed view of previous interactions, and samples an action $a_t \sim \pi_\theta(\cdot \mid c_t)$. The action a_t may include reasoning tokens, a tool invocation, an internal context-management operation, or a final answer. When a tool is invoked, the environment returns an observation o_t; for final answers or internal operations, o_t may be null. We denote one interaction turn, the history before step t, and the full trajectory by

$$u_t = (a_t, o_t), \qquad H_t = (u_1, \ldots, u_{t-1}), \qquad \tau = (u_1, \ldots, u_T), \tag{1}$$

with sparse final outcome reward $R(\tau)$. For each prompt x, GRPO (Shao et al., 2024) samples a group of N complete trajectories $\{\tau^{(1)}, \ldots, \tau^{(N)}\}$. The group-relative advantage of the n-th trajectory is

$$A^{(n)} = \frac{R(\tau^{(n)}) - \mathrm{mean}(\{R(\tau^{(i)})\}_{i=1}^{N})}{\mathrm{std}(\{R(\tau^{(i)})\}_{i=1}^{N}) + \epsilon}. \tag{2}$$

We use a simplified policy-gradient skeleton and suppress implementation-specific surrogate terms, such as old-policy ratios, clipping/gating, token normalization, and optional KL penalties, since our focus is how context

Table 1: Context-managed rollout strategies under the $(\mathcal{M}, \Phi)$ interface. Traceability indicates whether reconstructed context remains addressable at the level of original environment turns.

Strategy	**Managed History**	**Context Reconstruction**	**Traceability**	**Examples**
Append-only	$H_{K_{j-1}}$	$H_{K_{j-1}} \oplus H^{\mathrm{loc}}_{j,t}$	Explicit	Vanilla prompting
Truncation	Recent suffix or $\emptyset$	$H^{\mathrm{loc}}_{j,t}$	Lost	Sliding window
Recursive folding	$z_j = C(z_{j-1}, \sigma_{j-1})$	$\mathrm{Render}(z_j) \oplus H^{\mathrm{loc}}_{j,t}$	Collapsed	SUPO, MemAgent
Pruning / omission	$\mathcal{C}_j \subseteq H_{K_{j-1}}$	$\mathrm{Render}(\mathcal{C}_j, H^{\mathrm{loc}}_{j,t}; B)$	Partial	Agent-Omit
Memory actions	$\mathcal{B}_j$	$\mathrm{Render}(\mathcal{B}_j, H^{\mathrm{loc}}_{j,t}; B)$	Indirect	MemAct, Memory-R1
Selective turn memory	$M_j = \{e_i\}_{i \le K_{j-1}}$	$\mathrm{Render}(M_j[\widehat{\mathcal{I}}^{\mathrm{sel}}_j]) \oplus \Phi_{\mathrm{local}}(H^{\mathrm{loc}}_{j,t}; B)$	**Source-indexed**	**Ours**

reconstruction interacts with the final outcome signal:

$$\mathcal{J}_{\mathrm{MT}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T_n} A^{(n)} \log \pi_\theta(a_t^{(n)} \mid c_t^{(n)}) \right]. \tag{3}$$

This simplified form is used only to make explicit the dependence on the policy context and the trajectory-level advantage. In token-level implementations, each action a_t is a generated token sequence, and $\log \pi_\theta(a_t \mid c_t)$ is expanded as the sum of token log probabilities, with all generated tokens sharing the same trajectory-level advantage $A^{(n)}$. We distinguish the complete interaction history H_t from the policy context c_t: H_t is the environment-side record of previous turns, whereas c_t is the bounded textual input, or more generally the effective conditioning, provided to the language model policy.

3.2 Context-Managed Multi-Turn Rollouts

In long-horizon tool-use tasks, the complete history H_t grows with the number of interaction turns and may exceed the context budget available during rollout and training. Let B denote the maximum allowed length of the policy context. A context-managed rollout requires $|c_t| \le B$ for every interaction step. We partition each trajectory $\tau^{(n)}$ into bounded-context segments with boundaries

$$0 = K_0^{(n)} < K_1^{(n)} < \cdots < K_{J_n}^{(n)} = T_n. \tag{4}$$

For the j-th segment, we denote the completed prefix before the segment and the local history within the segment before turn t as

$$H^{(n)}_{K^{(n)}_{j-1}} = (u_1^{(n)}, \ldots, u^{(n)}_{K^{(n)}_{j-1}}), \qquad H^{(n),\mathrm{loc}}_{j,t} = (u^{(n)}_{K^{(n)}_{j-1}+1}, \ldots, u^{(n)}_{t-1}), \tag{5}$$

where $K^{(n)}_{j-1} < t \le K^{(n)}_j$. A context manager maps the completed prefix into a managed history state and reconstructs a bounded policy context:

$$z_j^{(n)} = \mathcal{M}\left(H^{(n)}_{K^{(n)}_{j-1}}\right), \qquad c^{(n)}_{j,t} = x \oplus \Phi\left(z_j^{(n)}, H^{(n),\mathrm{loc}}_{j,t}; B\right). \tag{6}$$

Here $z_j^{(n)}$ may take different forms, such as retained history, a summary, a compressed state, an edited memory, or a collection of memory units. The reconstruction function Φ combines this managed history with the local within-segment history under the budget B. We use $R(\tau)$ exclusively for rewards and reserve $H^{\mathrm{loc}}_{j,t}$ for local interaction history. The operator $\oplus$ denotes textual concatenation in prompt-based implementations, and more generally refers to the effective conditioning mechanism through which the task prompt and reconstructed history are provided to the policy.

We keep $\mathcal{M}$ and Φ abstract so that the interface can cover a broad range of context managers, including deterministic procedures, learned modules, and policy-generated context-management operations. When such operations are produced by the policy, their generated tokens are included in the trajectory and may receive credit in the same way as other action spans. For traceable methods, the reconstruction process may additionally expose metadata, such as the source indices of retained or selected history items, which can later be used for credit assignment. Substituting Eq. 6 into the simplified multi-turn objective gives

$$\mathcal{J}_{\mathrm{CM}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{N} \sum_{n=1}^{N} \sum_{j=1}^{J_n} \sum_{t=K^{(n)}_{j-1}+1}^{K^{(n)}_j} A^{(n)} \log \pi_\theta \left(a_t^{(n)} \mid x \oplus \Phi\left(z_j^{(n)}, H^{(n),\mathrm{loc}}_{j,t}; B\right) \right) \right]. \tag{7}$$

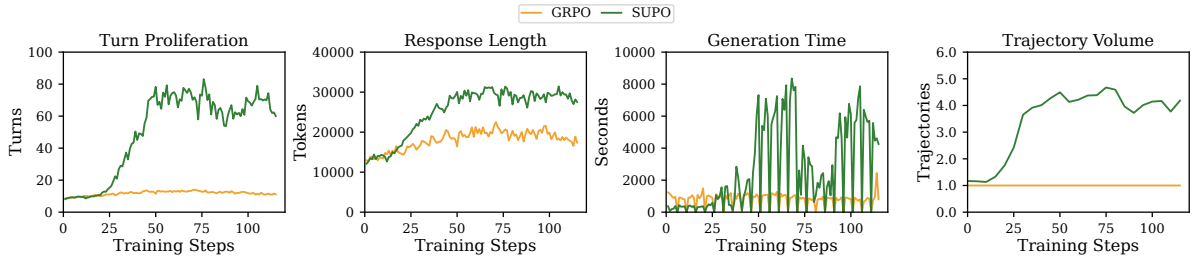


Figure 2: Training diagnostics on long-horizon search. Summarization-based context management enables longer rollouts, but leads to turn proliferation, longer responses, higher generation time, and increased trajectory volume.

This expression preserves the trajectory-level learning signal while making the policy input depend explicitly on the context-management mechanism. Different long-context strategies can thus be represented by different choices of $\mathcal{M}$ and Φ. The interface abstracts the effective conditioning available to the policy, rather than implementation details of a specific context manager. It also allows us to examine whether reconstructed information remains source-addressable for policy learning.

4 Method

4.1 Motivation: Context Reconstruction and Traceability

A context-reconstruction view. Eq. 6 defines a bounded-context rollout through two operations: a context manager $\mathcal{M}$ represents the completed prefix, and a reconstruction function Φ renders this representation into the policy context. This interface separates context efficiency for acting from source-level traceability for learning. A method may retain enough information for future decisions, but this does not imply that sparse outcome rewards can be routed back to the historical turns that supplied the useful evidence. Append-only prompting preserves explicit traceability but is not context efficient; truncation satisfies the budget by discarding distant history; and recursive folding compresses the prefix into a collapsed summary, which no longer directly exposes the original turns that contributed to later decisions. More generally, pruning, omission, and memory-action methods expose context-management decisions, but do not necessarily provide a shared source-indexed reconstruction trace for learning. Table 1 summarizes this source-traceability distinction under the $(\mathcal{M}, \Phi)$ interface.

Rolling summarization as a diagnostic. Rolling summarization illustrates the practical tension behind this distinction. By repeatedly folding old history into a summary, it enables longer bounded-context rollouts, but the resulting context is no longer directly addressable at the level of original environment turns. Figure 2 shows that this collapsed-history baseline can lead to rapid growth in tool-use turns, response length, generation time, and trajectory volume. These diagnostics do not prove that turn proliferation is caused solely by lost provenance; rather, they reveal an empirical failure mode of collapsed-history context management: longer rollouts may improve access to distant information, but can also encourage redundant search and increasingly expensive trajectories. Together with the source-traceability gap above, this motivates a context manager that prunes history for bounded-context acting while preserving explicit reconstruction traces for credit assignment.

4.2 Prune to Act: Selective Turn-Memory Reconstruction

We instantiate this source-indexed reconstruction principle in ECHO. The goal is to construct bounded policy contexts without collapsing the distant prefix into a single untraceable state. ECHO separates local turn compression from global context reconstruction: each completed turn is compressed independently into a source-indexed memory clue, while the active context is reconstructed by selecting from the resulting memory set.

Source-indexed turn memory. After each completed tool-use turn $u_i = (a_i, o_i)$, ECHO generates a compact local finding

$$s_i \sim \pi_\theta(\cdot \mid c_i^{\text{sum}}), \qquad m_i = \text{parse}(s_i), \tag{8}$$

where c_i^{sum} contains the latest turn u_i and a summary instruction x_{sum} that asks the model to summarize only that turn. The memory record is

$$e_i = (i, \alpha_i, m_i), \qquad M_j = \mathcal{M}(H_{K_{j-1}}) = \{e_i\}_{i \le K_{j-1}}. \tag{9}$$

Here i is the source-turn index, α_i is a compact rendering of the action or tool call, and m_i is the parsed last-turn finding. The finding is used as a lightweight clue for later selection and reconstruction, while the source index keeps the memory addressable. Unlike a rolling summary, M_j is a non-collapsing set: each memory remains individually linked to its original turn even after the raw observation falls outside the active context.

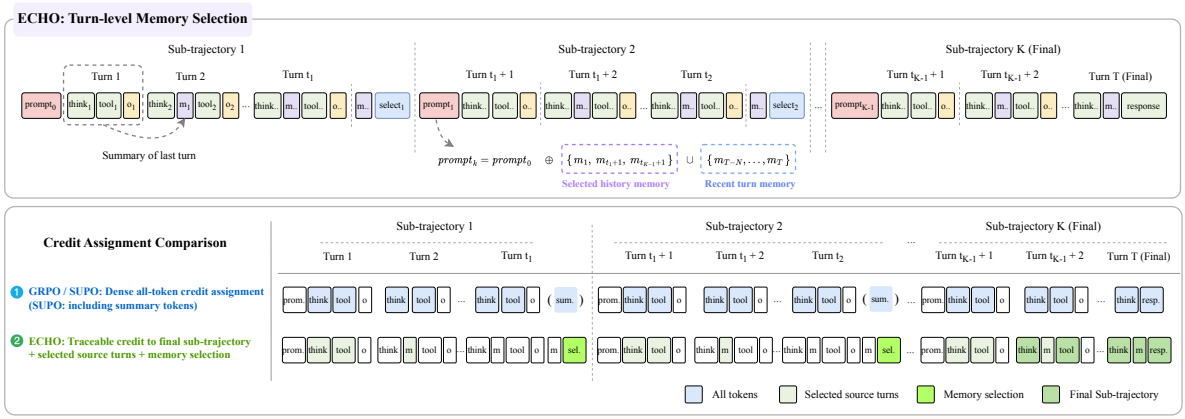


Figure 3: **Overview of ECHO.** ECHO stores completed turns as source-indexed memories, selects useful memories for bounded context reconstruction, and reuses the same source trace for credit assignment.

Autoregressive memory selection. Let S cap model-selected turns and K denote the latest turns retained automatically. At a compression boundary before segment j, let H_j^{bd} be the bounded local state available there. ECHO asks the policy to select historical memories useful for continuing the task:

$$c_j^{\mathrm{sel}} = x \oplus \Phi_{\mathrm{local}}(H_j^{\mathrm{bd}}; B) \oplus \mathrm{render}_{\mathrm{list}}(M_j) \oplus x_{\mathrm{sel}}, \quad a_j^{\mathrm{sel}} \sim \pi_\theta\big(\cdot \mid c_j^{\mathrm{sel}}\big), \quad \widehat{\mathcal{I}}_j^{\mathrm{sel}} = \rho_{B,S,K}\big(\mathrm{parse}(a_j^{\mathrm{sel}}), M_j\big). \tag{10}$$

The selection instruction x_{sel} asks the model to output source indices of reusable evidence, constraints, failed attempts, or planned next actions. The selection context abstracts the implementation: at a compression boundary, the selection instruction is appended to the current bounded segment state together with the rendered source-indexed memory list. The repair operator $\rho_{B,S,K}$ removes malformed or out-of-range indices, deduplicates selections, keeps at most S model-selected turns, merges them with the latest K turns retained automatically, and enforces the context-budget constraint.

Bounded context reconstruction. Selected source indices determine the memory used for reconstruction:

$$M_j[\widehat{\mathcal{I}}_j^{\mathrm{sel}}] = \{e_i \in M_j \mid i \in \widehat{\mathcal{I}}_j^{\mathrm{sel}}\}. \tag{11}$$

For any turn t in segment j, ECHO reconstructs the policy context as

$$c_{j,t} = x \oplus \mathrm{render}\Big(M_j[\widehat{\mathcal{I}}_j^{\mathrm{sel}}]\Big) \oplus \Phi_{\mathrm{local}}(H_{j,t}^{\mathrm{loc}}; B), \qquad |c_{j,t}| \le B. \tag{12}$$

Here Φ_{local} renders the recent within-segment history, possibly as a budgeted suffix. Thus, ECHO prunes distant history for acting while preserving an explicit provenance set $\widehat{\mathcal{I}}_j^{\mathrm{sel}}$. These selected source indices are the trace used for learning in Section 4.3.

4.3 Trace to Learn: Provenance-Guided Credit Assignment

The source indices selected for context reconstruction also define a credit route. As illustrated in Figure 3, GRPO and SUPO apply the trajectory advantage densely to generated tokens in the rollout; in SUPO this also includes tokens used to produce rolling summaries. Such dense assignment does not distinguish evidence turns from redundant searches once the final outcome reward is observed. ECHO instead first identifies credit tokens from the reconstruction trace, and only then attaches an outcome-dependent advantage to those tokens.

For trajectory n, let $\mathcal{I}_{\mathrm{src}}^{(n)} = \widehat{\mathcal{I}}_{J_n}^{(n),\mathrm{sel}}$ be the source turns selected into the final reconstructed context. Let q index generated response-token positions across the saved rollout segments. For each token position, $g_q^{(n)} \in \{0,1\}$ indicates whether the token is in the final response segment, $d_q^{(n)}$ is the source turn id of a normal assistant/action token, $f_q^{(n)}$ is the source turn id of a generated last-turn finding token, and $b_q^{(n)} \in \{0,1\}$ indicates whether the token belongs to a generated memory-selection span. ECHO constructs a token-level hard credit mask

$$\mu_q^{(n)} = \mathbf{1}\Big\{g_q^{(n)} = 1 \ \vee \ d_q^{(n)} \in \mathcal{I}_{\mathrm{src}}^{(n)} \ \vee \ f_q^{(n)} \in \mathcal{I}_{\mathrm{src}}^{(n)} \ \vee \ b_q^{(n)} = 1\Big\}, \tag{13}$$

which marks final-segment tokens, action tokens from source turns selected into the final reconstructed context, their last-turn finding tokens, and all generated memory-selection spans as credit tokens. Here $\mathcal{I}_{\mathrm{src}}^{(n)}$ defines a practical final-trace approximation: source-token credit is assigned only to turns selected into the final reconstructed

context, rather than to the full causal chain of all intermediate contexts. In contrast, selection-action credit is assigned to all memory-selection spans in positive-advantage rollouts, since these actions construct the available memory set during the trajectory. We then route an outcome-dependent advantage along this provenance trace. For binary verifier rewards $R(\tau) \in \{0, 1\}$, the credit term is

$$\widetilde{A}_q^{(n)} = A_+^{(n)} \mu_q^{(n)}, \qquad A_+^{(n)} = \max(A^{(n)}, 0), \tag{14}$$

where $A^{(n)}$ is the group-relative rollout advantage computed over valid rollouts for the same prompt. The choice of the positive part $A_+^{(n)}$ follows directly from what the selection trace means in each case.

For a correct rollout, the verified final answer makes the source turns selected into the final context a practical proxy for the historical evidence exposed to a successful decision. This trace routes positive credit to reused evidence paths, while screening off redundant searches and stale intermediate work not selected in the final reconstruction. **For an incorrect rollout**, the trace is not reliable: selected turns may be useful but misused, irrelevant, or misleading. Routing credit through such traces would introduce noisy supervision, so ECHO assigns traceable credit only to rollouts with positive group-relative advantage. Taking the positive part implements this rule: below-baseline and tied rollouts, including all-correct or all-incorrect groups, produce zero traceable update. Dense all-token credit is used only in the w/o Traceable CA ablation. The per-trajectory ECHO surrogate is

$$\mathcal{L}_{\text{ECHO}}^{(n)}(\theta) = \sum_q \widetilde{A}_q^{(n)} \log \pi_\theta(y_q^{(n)} \mid y_{<q}^{(n)}, c_q^{(n)}). \tag{15}$$

The overall objective averages Eq. 15 over prompts and sampled trajectory groups. As in Section 3, we suppress implementation-specific surrogate terms such as policy ratios, clipping or gating, token normalization, and optional KL penalties. Thus, ECHO routes outcome credit through the same source-indexed trace used to reconstruct the bounded context. Instead of assigning positive outcome credit to every generated token, it updates the final-segment tokens, the tokens attached to selected historical source turns, their memory clues, and the selection actions that exposed those memories to later decisions.

5 Experiment

5.1 Experimental Setup

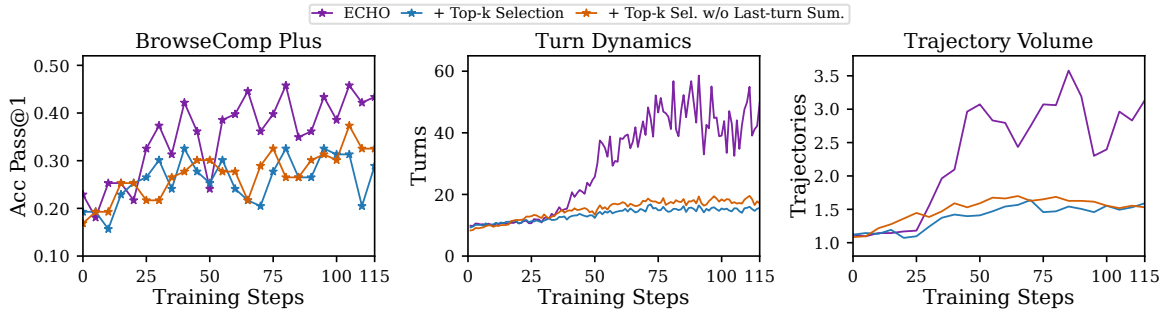
Benchmark and methods. We perform our primary evaluation on BrowseComp-Plus, a long-horizon tool-use QA benchmark requiring iterative search, evidence inspection, and answer verification. The training set contains 747 examples, and held-out validation contains 83 examples. Our main policy backbone is Qwen3-32B-Instruct, and we also evaluate transferability to the sparse MoE backbone Qwen3-30B-A3B-Instruct. We compare GRPO (Shao et al., 2024), SUPO-style rolling summarization (Lu et al., 2025), and ECHO under the same tool environment and verifier.

Implementation details. All methods use a 32k-token context budget, sample 8 rollouts per prompt, and train with the same GRPO-style outcome objective. For SUPO and ECHO, compression is triggered when the working context reaches this budget. We allow at most 5 compression rounds per rollout; together with the initial 32k segment, this gives an effective interaction budget of up to $6 \times 32\text{k} = 192\text{k}$ tokens before a rollout is treated as unsuccessful. ECHO summarizes each completed turn into a compact finding, retains the latest 3 turns automatically, and lets the policy select up to 8 additional historical turns during reconstruction. We report held-out pass@1 as the primary metric and track average turns, trajectory volume, response length, and generation time to characterize rollout behavior. Additional details are provided in Appendix C.1.

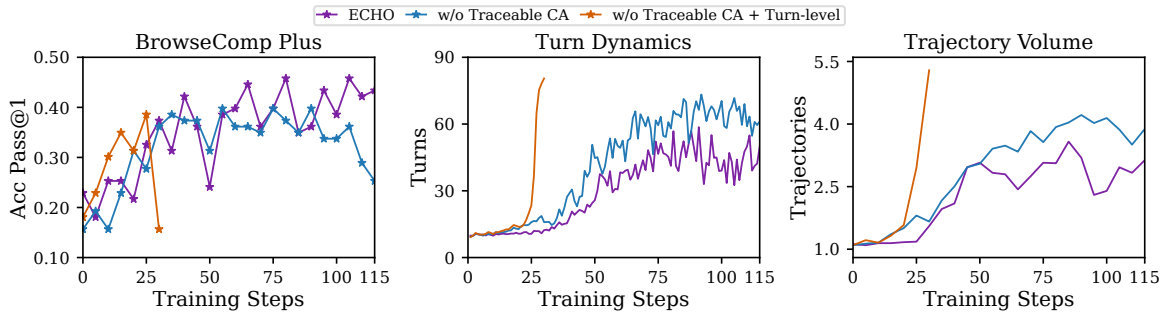
Zero-shot evaluation benchmarks. To assess out-of-domain generalization, we compare different methods without any additional tuning across three diverse domains: Multi-Objective QA spanning 2–16 objectives (Zhang et al., 2025), Code Generation on CodeGym (Du et al., 2025) and LoCoBench-Agent (Qiu et al., 2025), and Deep Information Seeking on GAIA (Mialon et al., 2024), HLE (Phan et al., 2025), and Frames (Krishna et al., 2025). Detailed dataset descriptions are deferred to Appendix C.2.

5.2 Main Results

Figure 1 compares training dynamics under an identical backbone, verifier, rollout budget, and sampling configuration. ECHO achieves a held-out accuracy of 43.4%, substantially outperforming both GRPO (28.9%) and SUPO (36.1%). GRPO keeps rollouts short, with 11.2 turns per rollout at the final step, but its accuracy plateaus early, suggesting that limited context exploration constrains long-horizon search. SUPO improves over GRPO by using rolling summaries to extend rollouts, but this gain comes with rapid turn proliferation: its final turn count reaches 62.5, with an 85.5% trajectory split rate and 4.18 trajectories per rollout on average. ECHO bridges this trade-off. It surpasses SUPO in accuracy while maintaining a more efficient rollout profile, ending with 45.3 turns,



(a) Memory component ablation.



(b) Credit assignment ablation.

Figure 4: Ablation study on BrowseComp-Plus. (a) Learned source selection in ECHO outperforms semantic top-k retrieval, while compact last-turn findings perform comparably to full observations under the same retrieval rule. (b) Removing traceable credit routing reduces accuracy and stability, while turn-level IS further induces rapid turn growth and lower accuracy.

a 57.8% trajectory split rate, and 3.13 trajectories per rollout on average. These results support our core hypothesis that source-indexed reconstruction and traceable credit routing enhance the utility of long-horizon search, rather than merely encouraging the policy to generate longer, redundant rollouts.

5.3 Ablation Study

Memory Component Ablation. We conduct ablations on two memory components of ECHO: memory selection and retrieved memory format. Specifically, we replace learned selection with static semantic top-k retrieval, and compact last-turn findings with full tool observations under the same retrieval protocol. Figure 4a shows that learned selection is the main contributor to performance. Although semantic top-k retrieval keeps rollouts compact, it yields substantially lower accuracy than ECHO, suggesting that useful historical information depends on the current search state rather than static semantic similarity alone. Moreover, full observations bring no clear improvement over compact findings, indicating that ECHO's gains primarily come from policy-driven source selection rather than more verbose retrieved content.

Credit Assignment Ablation. We next ablate credit assignment while keeping ECHO's reconstruction unchanged. ECHO w/o Traceable CA removes the traceable credit mask and falls back to a standard GRPO-style objective, assigning each rollout-level advantage to all generated tokens. ECHO w/o Traceable CA & Turn-level IS further replaces token-level importance-sampling ratios with turn-level ratios, computed by the geometric mean within each turn. As shown in Figure 4b, dense credit assignment reduces accuracy and stability: final outcome rewards are spread over useful evidence and redundant search, while turn-level IS treats each search or open-page turn as a coarse macro unit and can reinforce long continuation patterns. In contrast, ECHO routes credit only through final answers, selected source turns, and selection actions, better aligning updates with reused evidence.

5.4 Zero-shot Generalization Comparison

Table 2 evaluates whether the learned context-management behavior transfers beyond BrowseComp-Plus. On Qwen3-32B-Instruct, ECHO achieves the best average score of 40.2%, outperforming GRPO at 33.6% and SUPO at 34.8%. The gains are most pronounced in evidence-heavy settings such as 16-objective QA and Frames, where retaining addressable source evidence is especially useful. The same pattern holds under the MoE backbone, where ECHO reaches an average score of 30.5%, compared with 26.9% for SUPO. Moreover, the zero-shot

Table 2: Zero-shot generalization results across various benchmarks. **Bold** denotes the best performance and underlined denotes the second-best. CA denotes credit assignment, Traceable CA denotes source-indexed credit routing, and Turn-level IS denotes importance-sampling ratios aggregated at the turn level.

Method	Multi-Objective QA					Code Generation		Deep Information Seeking			Avg.
	2-obj.	4-obj.	8-obj.	16-obj.	Avg.	CodeGym	LoCoBench-Agent	GAIA	HLE	Frames	
Backbone: Qwen3-32B-Instruct											
GRPO	38.6	39.8	35.8	29.0	35.8	32.8	67.7	25.2	8.8	24.8	33.6
SUPO	40.9	36.4	36.4	34.7	37.1	35.4	68.1	25.2	9.2	26.8	34.8
ECHO w/ Top-K retrieval	**47.7**	42.0	39.2	27.8	39.2	40.7	69.3	**29.1**	10.6	37.3	38.2
ECHO w/ Top-K retrieval & w/o turn summary	40.9	44.3	35.8	35.5	39.1	40.3	69.5	23.3	10.0	31.3	36.8
ECHO w/o traceable CA	45.5	42.0	39.2	22.7	37.4	38.1	68.2	25.2	8.8	30.8	35.6
ECHO w/o Traceable CA & Turn-level IS	45.5	**47.7**	35.2	27.0	38.8	34.6	70.1	23.3	9.4	32.2	36.1
ECHO	**47.7**	45.5	**41.5**	**36.1**	**42.7**	**41.4**	**70.4**	**29.1**	**11.4**	**39.1**	**40.2**
Backbone: Qwen3-30B-A3B-Instruct											
GRPO	27.3	26.1	27.3	16.2	24.2	20.3	65.7	23.3	7.8	19.1	25.9
SUPO	25.0	30.7	27.3	18.2	25.3	27.3	65.1	23.3	8.0	17.0	26.9
ECHO	**34.1**	**36.4**	**30.1**	**18.8**	**29.9**	**29.7**	**66.8**	**24.3**	**9.2**	**25.0**	**30.5**

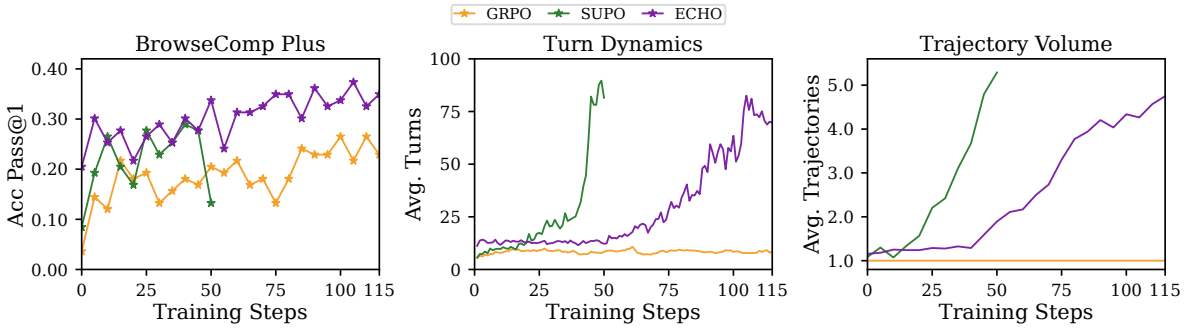


Figure 5: Training dynamics on BrowseComp-Plus with the Qwen3-30B-A3B-Instruct MoE backbone. SUPO is shown up to step 50, where the run enters a collapsed regime with sharply degraded accuracy and inflated average turns and trajectory volume.

ablations are consistent with the training curves: replacing learned selection with semantic top-k retrieval lowers the average score to 38.2%, while removing traceable credit further reduces it to 35.6%. These results suggest that both policy-driven memory selection and provenance-guided updates contribute to transfer, rather than merely improving performance on the training benchmark.

5.5 Robustness to Backbone Architecture

We repeat the BrowseComp-Plus training comparison with the sparse Qwen3-30B-A3B-Instruct MoE backbone, while keeping the tool environment, rollout budget, and evaluation protocol unchanged. This experiment tests whether the same source-indexed reconstruction and traceable-credit mechanisms remain effective under a different policy architecture. Figure 5 shows a similar method ranking to the dense-backbone setting. GRPO keeps turn counts lower but reaches only 22.9% accuracy. SUPO initially benefits from rolling summaries, but its turn count and trajectory volume grow rapidly as rollouts increasingly split and extend during training; by step 50, the resulting overlong trajectories reduce its accuracy to 13.3%. In contrast, ECHO remains stable after this point and finishes at around 35.0% accuracy. Although its turn count also increases, the additional search remains useful rather than merely inflating trajectory volume. Together with the MoE zero-shot results in Table 2, these findings suggest that the effectiveness of ECHO extends beyond the dense backbone.

6 Conclusion

We presented ECHO, a selective turn-memory framework for training long-horizon language agents under bounded contexts. The central idea is to make context reconstruction serve both acting and learning. Instead of collapsing distant history into an untraceable summary, ECHO stores each completed turn as a source-indexed memory, reconstructs compact contexts by selecting useful memories, and reuses the same source trace to route positive outcome credit to the final answer, reused evidence turns, memory findings, and selection actions. This design aligns the evidence retained for future decisions with the evidence reinforced after successful rollouts. On BrowseComp-Plus, ECHO improves held-out accuracy over GRPO and the rolling-summary baseline SUPO

while using fewer turns and lower trajectory volume than SUPO. Ablations show that both learned source selection and traceable credit routing are necessary for these gains, and zero-shot evaluations across multi-objective QA, code generation, and deep information-seeking benchmarks show that source-indexed reconstruction transfers beyond the training setting. These results suggest that preserving provenance during context management is a practical way to make long-horizon search more useful, not merely longer.

7 LIMITATIONS

We acknowledge two main limitations of the current study. First, ECHO adopts a *final-trace* approximation for credit assignment: it routes credit to the source turns selected into the final reconstructed context, but does not recursively follow the full dependency chain that produced those turns. A turn selected at the final step may itself have depended on earlier turns that the final trace no longer credits. This approximation is sufficient when the supporting evidence is shallow, but for very long-horizon tasks with deep nested dependencies it may undercover the historical turns that ultimately enabled success. Resolving such dependencies recursively would yield more complete credit at the cost of additional provenance tracking, which we leave to future work. Second, our experiments mainly focus on text-based tool-use agents. Although ECHO shows strong zero-shot transfer across several domains, broader validation on GUI, embodied, and multi-agent settings remains future work. In addition, source-indexed memory selection introduces extra computation, and more efficient memory indexing and selection strategies could further reduce training and rollout overhead.

REFERENCES

Zijian Chen, Xueguang Ma, Shengyao Zhuang, Ping Nie, Kai Zou, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, Sahel Sharifymoghaddam, Yanxi Li, Haoran Hong, Xinyu Shi, Xuye Liu, Nandan Thakur, Crystina Zhang, Luyu Gao, Wenhu Chen, and Jimmy Lin. BrowseComp-Plus: A more fair and transparent evaluation benchmark of deep-research agent. *arXiv preprint arXiv:2508.06600*, 2025.

Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.

Weihua Du, Hailei Gong, Zhan Ling, Kang Liu, Lingfeng Shen, Xuesong Yao, Yufei Xu, Dingyuan Shi, Yiming Yang, and Jiecao Chen. Generalizable end-to-end tool-use RL with synthetic CodeGym. *arXiv preprint arXiv:2509.17325*, 2025.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020.

Mengkang Hu, Tianxing Chen, Qiguang Chen, Yao Mu, Wenqi Shao, and Ping Luo. Hiagent: Hierarchical working memory management for solving long-horizon agent tasks with large language model. *arXiv preprint arXiv:2408.09559*, 2024.

Yuyang Hu, Hongjin Qian, Shuting Wang, Jiongnan Liu, Ziliang Zhao, Jiejun Tan, Zheng Liu, and Zhicheng Dou. SAM: State-adaptive memory for long-horizon reasoning agent. *arXiv preprint arXiv:2605.24468*, 2026.

Bowen Jin, Hansi Zeng, Zhenrui Yue, Wang Dong, Hamed Zamani, and Jiawei Han. Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516*, 2025.

Minki Kang, Wei-Ning Chen, Dongge Han, Huseyin A. Inan, Lukas Wutschitz, Yanzhi Chen, Robert Sim, and Saravan Rajmohan. Acon: Optimizing context compression for long-horizon llm agents. *arXiv preprint arXiv:2510.00615*, 2025.

Satyapriya Krishna, Kalpesh Krishna, Anhad Mohananey, Steven Schwarcz, Adam Stambler, Shyam Upadhyay, and Manaal Faruqui. Fact, fetch, and reason: A unified evaluation of retrieval-augmented generation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 4745–4759, 2025.

Ruoran Li, Xinghua Zhang, Haiyang Yu, Shitong Duan, Xiang Li, Wenxin Xiang, Chonghua Liao, Xudong Guo, Yongbin Li, and Jinli Suo. MemPO: Self-memory policy optimization for long-horizon agents. *arXiv preprint arXiv:2603.00680*, 2026.

Xuefeng Li, Haoyang Zou, and Pengfei Liu. ToRL: Scaling tool-integrated RL. *arXiv preprint arXiv:2503.23383*, 2025.

Yucheng Li, Bo Dong, Frank Guerin, and Chenghua Lin. Compressing context to enhance inference efficiency of large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

Zongqian Li, Yinhong Liu, Yixuan Su, and Nigel Collier. Prompt compression for large language models: A survey. *arXiv preprint arXiv:2410.12388*, 2024.

Miao Lu, Weiwei Sun, Weihua Du, Zhan Ling, Xuesong Yao, Kang Liu, and Jiecao Chen. Scaling LLM multi-turn RL with end-to-end summarization-based context management. *arXiv preprint arXiv:2510.06727*, 2025.

Grégoire Mialon, Clémentine Fourrier, Thomas Wolf, Yann LeCun, and Thomas Scialom. Gaia: a benchmark for general ai assistants. In *International Conference on Learning Representations*, volume 2024, pp. 9025–9049, 2024.

Yansong Ning, Jun Fang, Naiqiang Tan, and Hao Liu. Agent-omit: Adaptive context omission for efficient llm agents. *arXiv preprint arXiv:2602.04284*, 2026. doi: 10.48550/arXiv.2602.04284.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

Jiangweizhi Peng, Yuanxin Liu, Ruida Zhou, Charles Fleming, Zhaoran Wang, Alfredo Garcia, and Mingyi Hong. Hiper: Hierarchical reinforcement learning with explicit credit assignment for large language model agents. *arXiv preprint arXiv:2602.16165*, 2026.

Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, Chen Bo Calvin Zhang, Mohamed Shaaban, John Ling, Sean Shi, et al. Humanity's last exam. *arXiv preprint arXiv:2501.14249*, 2025.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 5687–5711, 2023.

Cheng Qian, Emre Can Acikgoz, Qi He, Hongru Wang, Xiusi Chen, Dilek Hakkani-Tür, Gokhan Tur, and Heng Ji. ToolRL: Reward is all tool learning needs. *arXiv preprint arXiv:2504.13958*, 2025.

Zile Qiao, Guoxin Chen, Xuanzhong Chen, Donglei Yu, Wenbiao Yin, Xinyu Wang, Zhen Zhang, Baixuan Li, Huifeng Yin, Kuan Li, et al. WebResearcher: Unleashing unbounded reasoning capability in long-horizon agents. *arXiv preprint arXiv:2509.13309*, 2025.

Jielin Qiu, Zuxin Liu, Zhiwei Liu, Rithesh Murthy, Jianguo Zhang, Haolin Chen, Shiyu Wang, Ming Zhu, Liangwei Yang, Juntao Tan, et al. Locobench-agent: An interactive benchmark for llm agents in long-context software engineering. *arXiv preprint arXiv:2511.13998*, 2025.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in neural information processing systems*, 36:68539–68551, 2023.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Lianlei Shan, Shixian Luo, Zezhou Zhu, Yu Yuan, and Yong Wu. Cognitive memory in large language models. *arXiv preprint arXiv:2504.02441*, 2025.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Weizhou Shen, Chenliang Li, Fanqi Wan, Shengyi Liao, Shaopeng Lai, Bo Zhang, Yingcheng Shi, Yuning Wu, Gang Fu, Zhansheng Li, et al. QwenLong-CPRS: Towards ∞-LLMs with dynamic context optimization. *arXiv preprint arXiv:2505.18092*, 2025.

Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Ji-Rong Wen. R1-Searcher: Incentivizing the search capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2503.05592*, 2025.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.

Hui-Ze Tan, Xiao-Wen Yang, Hao Chen, Jie-Jing Shao, Yi Wen, Yuteng Shen, Weihong Luo, Xiku Du, Lan-Zhe Guo, and Yu-Feng Li. Hindsight credit assignment for long-horizon llm agents. *arXiv preprint arXiv:2603.08754*, 2026.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.

Xiangfeng Wang, Zaiyi Chen, Zheyong Xie, Tong Xu, Yongyi He, and Enhong Chen. In-context former: Lightning-fast compressing context for large language model. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.

Yu Wang and Xi Chen. MIRIX: Multi-agent memory system for LLM-based agents. *arXiv preprint arXiv:2507.07957*, 2025.

Zixuan Wang, Yuchen Yan, Hongxing Li, Teng Pan, Dingming Li, Ruiqing Zhang, Weiming Lu, Jun Xiao, Yueting Zhuang, and Yongliang Shen. Milestone-guided policy learning for long-horizon language agents. *arXiv preprint arXiv:2605.06078*, 2026.

Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. BrowseComp: A simple yet challenging benchmark for browsing agents. *arXiv preprint arXiv:2504.12516*, 2025.

Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8:229–256, 1992.

Xixi Wu, Kuan Li, Yida Zhao, Liwen Zhang, Litu Ou, Huifeng Yin, Zhongwang Zhang, Yong Jiang, Pengjun Xie, Fei Huang, Minhao Cheng, Shuai Wang, Hong Cheng, and Jingren Zhou. Resum: Unlocking long-horizon search intelligence via context summarization. *arXiv preprint arXiv:2509.13313*, 2025.

Yating Wu, Yuhao Zhang, Sayan Ghosh, Sourya Basu, Anoop Deoras, Jun Huan, and Gaurav Gupta. Contextweaver: Selective and dependency-structured memory construction for llm agents. *arXiv preprint arXiv:2604.23069*, 2026.

Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-MEM: Agentic memory for LLM agents. *arXiv preprint arXiv:2502.12110*, 2025.

Yang Xu, Yunlong Feng, Honglin Mu, Yutai Hou, Yitong Li, Xinghao Wang, Wanjun Zhong, Zhongyang Li, Dandan Tu, Qingfu Zhu, et al. Concise and precise context compression for tool-using language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 16430–16441, 2024.

Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Hinrich Schütze, Volker Tresp, and Yunpu Ma. Memory-R1: Enhancing large language model agents to manage and utilize memories via reinforcement learning. *arXiv preprint arXiv:2508.19828*, 2025.

Chenxiao Yang, Nathan Srebro, David McAllester, and Zhiyuan Li. PENCIL: Long thoughts with short memory. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*, pp. 2369–2380, 2018.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.

Hongli Yu, Tinghong Chen, Jiangtao Feng, Jiangjie Chen, Weinan Dai, Qiying Yu, Ya-Qin Zhang, Wei-Ying Ma, Jingjing Liu, Mingxuan Wang, and Hao Zhou. MemAgent: Reshaping long-context LLM with multi-conv RL-based memory agent. In *International Conference on Learning Representations*, 2026.

Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, et al. DAPO: An open-source LLM reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.

Siliang Zeng et al. Reinforcing multi-turn reasoning in llm agents via turn-level credit assignment. *OpenReview*, 2025.

Chenchen Zhang. From reasoning to agentic: Credit assignment in reinforcement learning for large language models. *arXiv preprint arXiv:2604.09459*, 2026.

Yuxiang Zhang, Jiangming Shu, Ye Ma, Xueyuan Lin, Shangxi Wu, and Jitao Sang. Memory as action: Autonomous context curation for long-horizon agentic tasks. *arXiv preprint arXiv:2510.12635*, 2025.

Haotian Zhao, Songlin Zhou, Yuxin Zhang, Stephen S.-T. Yau, Wenyu Zhang, Lun Tian, Tianshu Zhu, Yifeng Huang, Yucheng Zeng, Jingnan Gu, Daxiang Dong, and Jianmin Wu. AEM: Adaptive entropy modulation for multi-turn agentic reinforcement learning. *arXiv preprint arXiv:2605.00425*, 2026.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.

Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye, Pengrui Lu, and Pengfei Liu. DeepResearcher: Scaling deep research via reinforcement learning in real-world environments. *arXiv preprint arXiv:2504.03160*, 2025.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

Zijian Zhou, Ao Qu, Zhaoxuan Wu, Sunghwan Kim, Alok Prakash, Daniela Rus, Jinhua Zhao, Bryan Kian Hsiang Low, and Paul Pu Liang. MEM1: Learning to synergize memory and reasoning for efficient long-horizon agents. *arXiv preprint arXiv:2506.15841*, 2025.

A Context Reconstruction Strategies

Figure 6 compares context reconstruction strategies under bounded-context agentic RL. The top pipeline shows the rollout loop, where the policy generates tool calls, receives observations, and triggers reconstruction once history exceeds the budget. The lower panels show how different methods reconstruct the next policy context. Append-only prompting preserves all turns but quickly exceeds the budget, while truncation, pruning, recursive folding, and memory-action methods shorten context by removing, compressing, or editing history. Although these strategies enable long-horizon rollouts, they often weaken the link between reconstructed context and original evidence turns. In contrast, ECHO stores completed turns as source-indexed memories and selects relevant ones for reconstruction, preserving explicit provenance for traceable credit routing.

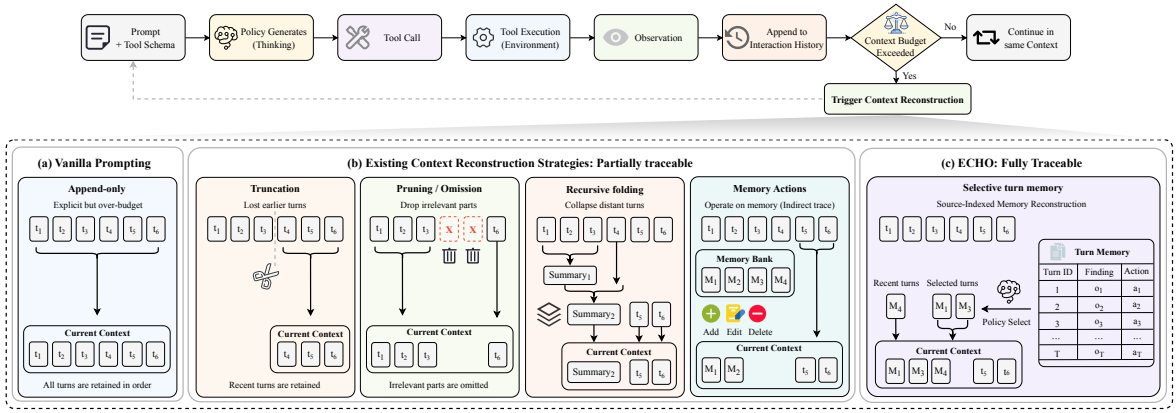


Figure 6: Existing context reconstruction strategies reduce context length but often lose full traceability. ECHO uses source-indexed selective memory to preserve evidence traceability for outcome-credit routing.

B ECHO Agent Loop

Algorithm 1 summarizes the rollout-time control flow used by ECHO. The agent maintains a working context C, a set of source-indexed memories M, and generated token segments $\mathcal{G}$. After each tool response, the policy writes a compact finding for the completed turn; when the context budget is exceeded, it selects relevant memory records, reconstructs a bounded continuation context, and records the provenance masks that will later be used for traceable credit assignment.

C Experimental Details

C.1 Implementation Setup

Each BrowseComp-Plus rollout has access to `search`, `open_page`, and `finish`. The search tool queries a local dense retrieval service using Qwen3-Embedding-8B and returns the top-5 documents by default. Each search result contains a document id, score metadata, and a snippet truncated to about 512 tokens. The `open_page` tool retrieves a document by `docid` and returns at most 16,000 characters. We allow at most 5 parallel tool calls per assistant turn, and `finish` terminates the rollout for BrowseComp-Plus verifier scoring.

All runs use the same fully asynchronous rollout-training pipeline on 4 nodes with 8 GPUs per node. Partial rollouts are enabled with staleness threshold 0.5. As described in Section 5.1, compression-based methods use a 32k-token context budget and at most 5 compression rounds, yielding an effective interaction budget of up to 192k tokens. Saved rollout segments share the rollout-level advantage; method-specific training masks determine which generated tokens receive nonzero updates.

C.2 Evaluation Benchmarks

In this section, we provide detailed descriptions of the evaluation protocols and dataset characteristics for the zero-shot generalization experiments.

Multi-Objective QA. To evaluate the agent's long-range reasoning and context management capabilities under varying cognitive loads, we construct a multi-objective QA benchmark (Zhang et al., 2025). This benchmark challenges the agent to resolve multiple independent sub-questions within a single extended rollout to synthesize

Algorithm 1 ECHO rollout-time agent loop

Require: Task prompt x, policy π_θ, tools $\mathcal{T}$, context budget B, recent-turn budget K, selection cap S

$C \leftarrow x$
$M \leftarrow \emptyset$; $G \leftarrow \emptyset$; $\mathcal{G} \leftarrow \emptyset$; $p \leftarrow \emptyset$; $i \leftarrow 0$
while rollout is not terminated **do**
  Sample $a_t \sim \pi_\theta(\cdot \mid C)$
  Append a_t to C and append its trainable tokens to G
  if $p \neq \emptyset$ **then**
    $(i_p, \alpha_{i_p}) \leftarrow p$
    $m_{i_p} \leftarrow \text{parse} * \text{sum}(a_t)$
    $M \leftarrow M \cup e * i_p = (i_p, \alpha_{i_p}, m_{i_p})$
    Tag the summary tokens in G as finding tokens for source turn i_p
    $p \leftarrow \emptyset$
  end if
  Parse tool calls from a_t
  if no tool call remains **then**
    Mark terminal-answer tokens in G as final
    $\mathcal{G} \leftarrow \mathcal{G} \cup G$; **break**
  end if
  $\alpha_i \leftarrow \text{render} * \text{act}(a_t)$
  Tag the non-summary action tokens in G as action tokens for source turn i
  Execute tool calls with $\mathcal{T}$ and obtain observations o_i
  Append observations o_i to C
  $p \leftarrow (i, \alpha_i)$; $i \leftarrow i + 1$
  Append the <sum_last_turn> hint to C
  if $|C| > B$ **then**
    $c_j^{\text{sel}} \leftarrow C \oplus \text{render} * \text{list}(M) \oplus x_{\text{sel}}$
    $a_j^{\text{sel}} \sim \pi_\theta(\cdot \mid c_j^{\text{sel}})$
    Append a_j^{sel} to G and tag its tokens as selection tokens
    $I \leftarrow \rho_{B,S,K}(\text{parse}(a_j^{\text{sel}}), M)$
    Record selected-turn, finding-token, and selection-token masks
    Mark G as non-final; $\mathcal{G} \leftarrow \mathcal{G} \cup G$
    $C \leftarrow x \oplus \text{render}(M[I]) \oplus x_{\text{cont}}$
    $G \leftarrow \emptyset$
  end if
end while
return $\mathcal{G}$ with credit-token masks

a final answer. We synthesize these multi-objective queries using seed instances from HotpotQA (Yang et al., 2018), 2WikiMultihopQA (Ho et al., 2020), Bamboogle (Press et al., 2023), and Musique (Trivedi et al., 2022). We scale the task difficulty by aggregating questions to form test sets requiring 2, 4, 8, and up to 16 distinct objectives. Performance is measured using an LLM-as-a-Judge (Zheng et al., 2023) protocol. The evaluator assesses the semantic consistency between the agent's final answer and the ground truth, computing the average success rate across all sub-objectives within a given multi-objective prompt.

Code Generation. We evaluate interactive programming capabilities using CodeGym (Du et al., 2025) and LoCoBench-Agent (Qiu et al., 2025). CodeGym frames tasks as synthetic interactive environments where agents invoke problem-specific APIs, together with `observe()` and `done()`, rather than writing raw code. Since many CodeGym tasks are either too easy or too difficult for meaningful comparison, we construct a medium-difficulty subset using the original `Qwen3-32B-Instruct` policy before agentic RL training. Specifically, we first run the base policy four times per instance, retain instances solved exactly once, and then randomly sample 128 examples from this candidate pool. This filtering is performed once before evaluating any trained method, and the same subset is used for all methods. For authentic software-engineering scenarios, we further evaluate on 128 Python instances from LoCoBench-Agent. These tasks operate on real codebases of up to 1 million tokens and require agents to manage cross-file dependencies, investigate bugs, and perform architectural refactoring with tools for file operations and semantic search.

Deep Information Seeking. Deep information seeking benchmarks measure the agent's proficiency in orchestrating complex reasoning, multi-step tool use, and long-horizon fact aggregation. We evaluate on GAIA (Mialon et al., 2024), which reflects real-world assistant tasks demanding rigorous execution and robust fact-checking. To test the limits of domain expertise, we use Humanity's Last Exam (HLE) (Phan et al., 2025), a frontier-level

benchmark featuring expert-vetted questions that require deep reasoning rather than surface-level internet retrieval. Finally, we incorporate Frames (Krishna et al., 2025) to assess the agent's capacity to resolve multi-hop queries, integrate conflicting evidence, and maintain critical factual anchors across extensive retrieval trajectories without experiencing history collapse.

D Prompt Details

D.1 Base Prompt for BrowseComp-Plus

System and User Prompt

We use the same BrowseComp-Plus tool-agent prompt setting as the SUPO baseline. The system and user prompts follow the same multi-step research protocol for query decomposition, iterative search, evidence synthesis, and final verification. The tool interface is formatted with the Hermes-style `<tool_call>...</tool_call>` protocol.

D.2 ECHO-Specific Prompts and Hints

ECHO Extra System Prompt

After receiving a tool/function response, your next assistant message must include exactly one `<sum_last_turn>...</sum_last_turn>` block before any new tool/function call.
The block should be one concise factual sentence summarizing only the latest tool/function result.
After the block, continue in the same assistant message with the next tool/function call, or call `finish` if the final answer is ready. Do not stop after the summary.

Sum Last Turn Hint

Briefly record the latest tool result in `<sum_last_turn>...</sum_last_turn>`, then continue with the next action.

Selection Prompt

System:
Your operational context is full. Select prior interaction turns that are necessary to continue solving the task.

Rules:

- Output exactly one `<selection>...</selection>` block.
- Do not call any function/tool in this turn.
- Do not answer the original task.
- Do not include `<think>`, tool calls, markdown fences, or text outside the selection tags.
- Select only turns that contain reusable evidence, constraints, failed attempts, or the next planned action.
- Prefer older turns that are not already covered by automatically retained recent turns.
- If no older turn is necessary, return an empty `<selection></selection>` block.

Valid turns:
{turn_list}

```
<selection>[zero or more lines, each formatted as turn_N: reason]
</selection>
```

Selection Hint

Valid selection indices: turn_0 $\sim$ turn_{n_hist-1}.
Select at most {selection_max_turns} historical turns; output an empty selection if the automatically retained recent turns are enough.
The latest turns turn_{recent_start} $\sim$ turn_{n_hist-1} will be retained automatically; select any additional older turns that are still needed.

In our BrowseComp-Plus experiments, `selection_max_turns = 8` and `echo_recent_turns = 3`.